

Abhirup Paul

Mobile-First Full Stack Engineer | React Native | REST APIs & Backend Systems

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Professional Summary

Mobile-First Full-Stack Engineer with hands-on, production experience building and scaling a high-performance, cross-platform **React Native** mobile application (**iOS & Android**) end-to-end – owning features from development through deployment, integrated with **REST APIs**, **GraphQL**, and **WebSocket** services. Comfortable contributing to backend development (**Python/FastAPI**, business logic, API design) and working closely with backend teams to design efficient, scalable APIs. Strong understanding of mobile app architecture, performance, debugging, and **system design**, with production experience ensuring app scalability and reliability in a live, product-based clinical environment used daily across two hospitals. Fast learner actively expanding into native **Android (Kotlin)**, **iOS (Swift)**, and **Flutter/Dart** to broaden cross-platform mobile expertise.

Core Competencies: Mobile App Development | React Native (iOS & Android) | REST & GraphQL APIs | Backend API Design (Python/FastAPI) | App Architecture, Performance & Debugging | System Design & Scalable Architecture | SQL & NoSQL Databases | CI/CD & Cloud Deployment (AWS)

Experience

Mobile & Full-Stack Engineer (R&D) **Jul 2025 – Present**
SCAN Lab, Indian Institute of Technology Bombay Mumbai, India

- Built and scaled **Neuroscan360**, a high-performance, patented enterprise **React Native** mobile application (**iOS & Android**) – owned the mobile feature lifecycle end-to-end from development through deployment for a clinical screening product live in two major Mumbai hospitals, used daily by multiple medical teams.
- Integrated and consumed **REST APIs** and **WebSocket (Socket.IO)** services for real-time multimodal data streaming; parsed **JSON** payloads across concurrent clinical pipelines while maintaining a low-latency, reactive UI; implemented native device sensor and camera integration.
- Owned mobile app architecture, performance tuning, and debugging – resolved cross-platform issues, optimized rendering, and built reusable **Redux**-based UI components (MVVM/MVC) to ensure the app remained scalable and reliable.
- Collaborated with backend engineers to design efficient **REST APIs** and contributed directly to backend development (**FastAPI**, Golang, **PostgreSQL**, business logic) supporting the mobile app; shipped via **CI/CD** (GitLab CI, GitHub Actions, **AWS**, Docker, Kubernetes) – improved reliability by 64% and cut costs by 30%.
- Applied **system design** principles to build scalable, maintainable services; integrated LLM APIs (Claude/OpenAI) and RAG pipelines via LangChain for AI-assisted workflows, authoring test cases and QA reports.

Freelance Backend Engineer **Dec 2024 – Feb 2025**
Independent, Remote (Startup-Style Engagement) Remote

- Designed and shipped a rule-based fraud-signal engine for real-time financial transactions in a fast-paced, product-focused engagement – owned **REST API** design, data modeling, JWT-secured endpoints, and event processing (**Python**, **FastAPI**, **PostgreSQL**, **Redis**, **AWS**).
- Built a **WebSocket** real-time alerting system with ledger-aware transaction integrity; maintained structured Markdown API contracts, test reports, and deployment logs throughout.

Projects

Event Ticketing Management System Python | FastAPI | PostgreSQL | Redis | WebSocket | Stripe | Docker | GitHub Actions

- Integrated Stripe **REST API** and parsed **JSON** webhook payloads for idempotent payment orchestration; implemented Redis distributed locks for race-free seat reservation.
- Authored and maintained Markdown KB (API schemas, changelogs, deployment runbooks); executed automated tests, documented anomalies, and shipped via **CI/CD** (GitHub Actions).

Real-Time Data Pipeline & Analytics Golang | Redis | WebSocket | REST APIs | Pub/Sub

- Built a high-throughput pipeline processing 10,000+ events/day at sub-100ms latency; applied structured logging and anomaly flagging.
- Authored version-tracked Markdown knowledge base (data schema, API endpoints, operational runbook).

Technical Skills

Mobile Development:	React Native (iOS & Android), Redux, Native Module Bridging, Device Sensors & Camera Integration, Cross-Platform Mobile Development, React.js, Next.js
Mobile Platforms (Growth Areas):	Native Android (Kotlin) , iOS (Swift) , Flutter/Dart – actively expanding cross-platform mobile expertise
Backend & APIs:	Python (FastAPI) , Golang, REST API Design, GraphQL , Microservices, JWT, WebSocket, Concurrency (goroutines/channels)
Databases:	PostgreSQL , MongoDB, Redis, Firebase, pgvector (SQL & NoSQL)
System Design & Architecture:	Scalable System Design, Microservices Architecture, App Performance & Reliability Engineering
DevOps & QA:	Docker, Kubernetes, GitLab CI, GitHub Actions, CI/CD, AWS , Unit Testing, Automated Test Pipelines, Code Review, Jira
AI / LLM Tooling:	LLM API Integration (Claude/OpenAI), RAG Pipelines, LangChain, Claude Code, Cursor

Education

B.Tech, Electronics & Computer Engineering **2022 – 2026**
GKV FET, Haridwar, Uttarakhand, India CGPA: 8.5 / 10.0
Research: SCAN Lab, IIT Bombay under Prof. Samir K. Maji – ML & DL for biomedical signal processing and neurological assessment.
Coursework: Machine Learning, NLP, Data Structures, Computer Networks, Database Systems.

Achievements

- Co-inventor – Neuroscan360 Patent (IIT Bombay):** Clinical mobile platform deployed in two major Mumbai hospitals, used daily for neurological assessments by multiple medical teams.
- Runner-up – MumbaiHacks 2024:** Built and presented an AI-powered solution in a competitive 24-hour hackathon.